Real-World Home Practice for Children's Speech Sound Disorders

An invitation to research partnership from SpeechLP

November 2026

SpeechLP records every scored articulation practice attempt children make at home, with timestamps, target sounds, and practice frequency. In a first look at 101 children who practiced the same sounds over two consecutive months, average app-scored accuracy rose from 69.4% to 74.9%. We are looking for university researchers to help answer what this data can and can't show, starting with two small, focused studies.

Why this matters

Home practice is a standard part of treatment for speech sound disorders, yet the field knows little about how much children actually practice or how practice relates to progress.

- **Home practice is standard but rarely measured.** SLPs commonly involve parents through home practice, but published surveys of SLPs haven't quantified how often it is assigned (Sugden et al., 2018; Tambyraja, 2020), and intervention studies seldom report what that practice involved: a review of the parent-involvement literature found most papers gave minimal detail on what parents did or how much they practiced (Sugden et al., 2016). ASHA's national outcomes data found that three times as many children with severe speech sound disorders made significant functional progress in intelligibility when therapy combined 21–30 sessions, scheduled twice weekly, with an SLP-directed home program, compared with children who received fewer sessions or whose plan lacked any of these elements (ASHA, 2024).
- **Dose is understudied.** How the number of practice trials and their frequency relate to outcomes is still an open question. In a randomized study of preschoolers, three sessions a week produced larger gains than one session a week at the same total intensity (Allen, 2013). Motor-learning research points to practice amount, distribution, variability, and feedback as key factors (Maas et al., 2008). Similarly, in a comparative pre-/post-treatment study of 37 children aged 32–54 months with childhood apraxia of speech, with stratified randomization of clinicians to treatment-intensity groups, twice-weekly motor speech treatment for 10

weeks outperformed once-weekly treatment on articulation and functional communication, though neither group improved in intelligibility (Namasivayam et al., 2015).

- **Early treatment matters.** Speech sound disorders affect an estimated 2.1–23% of children aged 4–6, and early speech sound difficulties are associated with later literacy problems (ASHA, n.d.). Children whose speech difficulties persist, or occur alongside language difficulties, are at greater risk for later literacy problems (Raitano et al., 2004; Peterson et al., 2009; Hayiou-Thomas et al., 2017). A 2025 meta-analysis found moderate reading deficits in children with speech sound difficulties, both concurrently (g = −0.58) and over time (g = −0.50), though the authors note possible publication bias (Walquist-Sørli et al., 2025). Treatment is effective at these ages: a randomized study of 54 preschool children found significant gains (Allen, 2013; see dose above), and a meta-analysis of controlled studies with children aged 3–18 found medium-to-high effects on phonological accuracy, with no significant moderators detected (Kunnari et al., 2024).
- **Carryover is the hard part.** Treatment moves from establishing a sound to generalizing it into conversation and maintaining it (ASHA, n.d.), and self-monitoring has long been studied as a way to speed that transfer (Shriberg & Kwiatkowski, 1990). Recent work shows carryover isn't automatic: when four children aged 4–6 used a speech-training app at home without supervision for three weeks, two improved slightly on trained words but not on untrained words, and the authors recommended more parental involvement and better in-game feedback (Strömbergsson et al., 2026). A meta-analysis of 12 treatment studies found that treating more complex sounds also improved untreated, simpler sounds (Maggu et al., 2021).
- **Automated scoring needs validation.** AI-assisted practice without a clinician improved listener-rated /ɹ/ in a single-case experimental study, using a system built and evaluated against clinician judgment (Benway & Preston, 2024). That agreement is what makes automated scores clinically meaningful. Expert listening has limits too: more experienced SLPs use the in-between points of a rating scale less often, and clinicians' intelligibility estimates aren't accurate or reliable enough on their own (Meyer & Munson, 2021; Allison et al., 2025). Namasivayam et al. (2025) argue for combining expert listening with instrumental measures, and describe automated speech recognition as holding great promise for assessing and supplementing treatment of speech sound disorders. In a Swedish study of 28 children aged 4–10, a speech recognition model rated single words on a five-level scale (from unidentifiable to correct, including subphonemic errors) within about 5 percentage points of an expert's own re-ratings on accuracy, though the expert was clearly better at telling the middle levels apart (Getman et al., 2022). The design has not been tested in English. Similar

models have shown promise in other languages: in 30 Korean children aged 3–7 with suspected speech sound disorders, an automated assessment showed potential compared with SLPs, though its performance varied by phoneme and word (Kim et al., 2025), and a model trained on 798 Mandarin-speaking preschoolers reached a 3.76% phoneme error rate, with clinical indices closely tracking age norms (Lin et al., 2024). None has been validated for English home practice.

SpeechLP's records speak directly to the first two gaps, and a partnership could address the other three.

The dataset at a glance

Every scored practice item is logged with the word attempted, sounds recognized, whether it was accepted, a timestamp, and session, across all 24 English consonants. The dataset is still growing.

- **First analysis (Jan–Oct 2026):** 101 children with two full months of data (55 aged 2–5, 33 aged 6–8; the other 13 were aged 9–12 or had no recorded age, counts withheld for privacy) and 17,675 scored practice cards.
- **SLP-assigned practice:** through SpeechTherapist.app, SLPs invite a caregiver, assign home practice by target sound, word position, and consonant blend, and track home sessions, words practiced, and accuracy overall and by sound.
- **Word preview:** before playing, a "Words in this game" panel shows the target sound and each word, with the target sound highlighted and a button to hear the word, and each practice card also plays a spoken model the child can replay. Studies can therefore distinguish imitated from spontaneous productions if model playback is logged.
- **Platforms:** children practice in the iPhone and iPad app or the web version; the word list (1,500+ words) spans ages 2–13, and the app is designed for ages 2–9.
- **Privacy:** speech recognition runs on the device where supported. Apple's speech recognition, used as a backup checker, may process audio on the device or on Apple's servers, subject to Apple's terms of service. SpeechLP never receives any audio. Transcribed text is stored as depersonalized data, with all direct and indirect personal identifiers removed or masked. An optional "Record a run" feature saves the child's voice on the family's own device.

Early findings

Among 101 children practicing the same sounds in two consecutive 30-day periods, app-scored accuracy rose from 69.4% to 74.9%, while average practice fell from 6.7 to 3.4 sessions and from 4.7 to 2.7 practice days per child.

- **Accuracy rose 5.5 percentage points** (95% CI 0.4 to 10.4). 54 children improved, 34 declined, and 13 were unchanged.
- **Most sounds moved up.** 16 of 21 reportable consonants had higher average accuracy in Month 2; /k/, /t/, and /z/ showed the clearest gains.
- **More practice did not clearly mean bigger gains.** Children with 5 or more first-month sessions gained 7.5 points and those with 1–4 gained 3.3, but the difference was not statistically reliable.

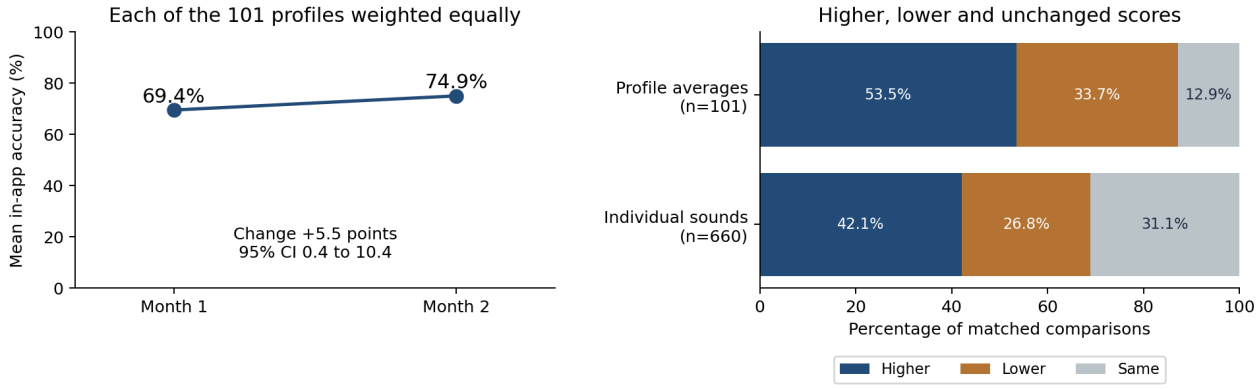


Figure. Mean in-app accuracy across the 101 profiles (left) and the share of profiles and individual sounds with higher, lower, or unchanged scores (right). From the full report (Silver et al., 2026).

What the data can't show yet. These are observational results with no comparison group. Children who kept practicing for two months may differ from those who stopped, and some change may reflect regression toward the mean. Accuracy is the app's own scoring, not yet checked against SLP judgment. Most children were aged 2–5, when many sounds are still emerging; most English consonants are acquired by age 5;0, with /ɹ/, /θ/, /ð/, and /ʒ/ last (Crowe & McLeod, 2020), so some gains may reflect normal development. Diagnosis and severity aren't recorded, though families connected to an SLP could supply them with consent, and in-app accuracy doesn't show carryover to untrained words or conversation. The studies below are designed to answer these questions.

Proposed studies

We suggest two starting studies; a partner could take on one or shape their own. In each, SpeechLP provides practice data, the app platform, engineering support, and recruitment through connected SLPs. The partner leads study design, ethics approval, and analysis. Together they follow two steps: first, validate the app's correct/incorrect scoring against SLPs, which few home-practice tools have done in English; then use the same platform to study the gradual, multidimensional signs of progress that correct/incorrect scoring misses. Validation results will also guide our ongoing fine-tuning of SpeechLP's speech recognition model. A recent model update improved recognition across consonants; further work aims to reduce reliance on Apple's speech recognition as a backup checker and to detect phonological processes.

Starter pilot (could support a co-authored ASHA Convention submission)

1. **Scoring validation.** Compare the app's scores with expert SLP ratings, acoustic measures, and parents' correct/incorrect judgments during home practice, across children aged 3–12 with a range of speech motor skill. This would replicate in English the human-versus-machine design of Getman et al. (2022), with several SLP raters instead of one and the paper's metrics (accuracy, unweighted average recall, mean absolute error), and add the parent comparison the authors name as their next step. The partner records study sessions under its ethics approval.

Starter study using existing data

2. **Home practice dose.** Test whether the amount, spacing, and accuracy of home practice predict gains, and how closely families complete SLP-assigned practice. Data access is de-identified and governed by a data use agreement agreed with the partner.

Longer-term directions. Once scoring is validated, the same platform could support larger studies designed together: randomized comparisons of feedback type and frequency within the app, and prospective outcome studies with independent and functional measures such as percentage of consonants correct (Shriberg et al., 1997) and the Intelligibility in Context Scale (McLeod et al., 2012), plus stimulability, self-monitoring, a comparison group, and follow-up of early literacy outcomes such as phonological awareness and decoding.

What partnering looks like

Partner researchers keep full scientific independence, including the right to publish whatever they find.

- **Independence.** Researchers control the analysis and can publish regardless of results. SpeechLP reviews manuscripts within 30 days only to protect participant privacy and to correct factual errors about the product, not to approve findings.
- **Agreements.** A data use agreement covers access, privacy, storage, and ownership of any models or methods a partner develops using SpeechLP data. Ethics approval runs through the partner's institutional review board.
- **Privacy.** SpeechLP complies with COPPA, FERPA, HIPAA, and PIPEDA. Data is shared de-identified under a data use agreement, with small groups suppressed. Terms for research data sharing are set with each partner in the data use agreement. Account and child identifiers never leave SpeechLP, and SpeechLP never receives any audio from the app; transcribed text is stored as depersonalized data. Study recordings are collected and held by the partner; SpeechLP receives audio only if participants separately and explicitly consent to share recordings for model development, as covered by the data use agreement.
- **Support we can offer.** App licenses for study participants, engineering time for data exports, and co-authorship for SpeechLP clinicians where contributions warrant it. Funding can be discussed once a specific study is scoped.
- **Transparency.** SpeechLP's cofounders have a financial interest in the app, and this is disclosed in all outputs.

Team and next step

SpeechLP was founded and is led by practicing speech-language pathologists.

- **Alissa Fox, M.S., CCC-SLP** — Cofounder and Co-CEO; SLP since 2004, treating children's speech sound disorders, motor speech delays, and language disorders in schools and via teletherapy, with a focus on assistive technology; Vice President for Schools at the Iowa Speech-Language-Hearing Association; licensed in Pennsylvania and Iowa.
- **Mari Gottlieb, M.A., CCC-SLP** — Cofounder and Co-CEO; private practice owner and SLP since 2017, treating children's speech sound and language disorders in schools, and cognitive-communication impairments during nearly six years at a brain injury and concussion clinic;

SPEAK OUT!-trained; brings a prior advertising career to family engagement and app design; licensed in Colorado and Washington.

- **Shail Silver** — Cofounder and Chairman.
- **Jabali Ndungu** — Head of Engineering; data and analysis support.
- **Matthew Siu** — Software Engineer; data and analysis support.
- **Cindy Cheng** — UX/UI Designer; designs the SpeechLP and SpeechTherapist.app interfaces, games, and characters.

Next step: a 30-minute call to hear about your research interests and see whether one of these studies, or another question, fits. Contact Alissa Fox at alissa@speechlp.com.

The full technical report, *Children's Articulation Practice in SpeechLP Across Two Months* (November 3, 2026; https://doi.org/10.5281/zenodo.23932485), is attached as a technical appendix.

References

Allen, M. M. (2013). Intervention efficacy and intensity for children with speech sound disorder. *Journal of Speech, Language, and Hearing Research*, *56*(3), 865–877. https://doi.org/10.1044/1092-4388(2012/11-0076)

Allison, K. M., Millington, M., & Grimm, A. (2025). Can we trust our ears? How accurate and reliable are speech-language pathologists' estimates of children's speech intelligibility? *American Journal of Speech-Language Pathology*, *34*(2), 853–867. https://doi.org/10.1044/2024_AJSLP-24-00247

American Speech-Language-Hearing Association. (n.d.). *Speech sound disorders: Articulation and phonology* [Practice portal]. https://www.asha.org/practice-portal/clinical-topics/articulation-and-phonology/

American Speech-Language-Hearing Association. (2024). *Dosage and home practice matters: Improving intelligibility in children with severe speech sound disorders* [National Outcomes Measurement System data snapshot]. https://www.asha.org/siteassets/noms/noms-speech-sound-disorders-data-snapshot.pdf

Benway, N. R., & Preston, J. L. (2024). Artificial intelligence–assisted speech therapy for /ɹ/: A single-case experimental study. *American Journal of Speech-Language Pathology*, *33*(5), 2461–2486. https://doi.org/10.1044/2024_AJSLP-23-00448

Crowe, K., & McLeod, S. (2020). Children's English consonant acquisition in the United States: A review. *American Journal of Speech-Language Pathology*, *29*(4), 2155–2169. https://doi.org/10.1044/2020_AJSLP-19-00168

Getman, Y., Al-Ghezi, R., Voskoboinik, E., Grósz, T., Kurimo, M., Salvi, G., Svendsen, T., & Strömbergsson, S. (2022). wav2vec2-based speech rating system for children with speech sound disorder. In *Proceedings of Interspeech 2022* (pp. 3618–3622). https://doi.org/10.21437/Interspeech.2022-10103

Hayiou-Thomas, M. E., Carroll, J. M., Leavett, R., Hulme, C., & Snowling, M. J. (2017). When does speech sound disorder matter for literacy? The role of disordered speech errors, co-occurring language impairment and family risk of dyslexia. *Journal of Child Psychology and Psychiatry*, *58*(2), 197–205. https://doi.org/10.1111/jcpp.12648

Kim, D. H., Jeong, J. W., Kang, D., Ahn, T., Hong, Y., Im, Y., Kim, J., Kim, M. J., & Jang, D.-H. (2025). Usefulness of automatic speech recognition assessment of children with speech sound disorders: Validation study. *Journal of Medical Internet Research*, *27*, e60520. https://doi.org/10.2196/60520

Kunnari, S., Sanduvete-Chaves, S., Chacon-Moscoso, S., Alves, D. C., Ozbič, M., Petinou, K., Tolonen, A.-K., Zajdó, K., Frizelle, P., Murphy, C.-A., Saldana, D., & Laasonen, M. (2024). Intervention studies with group design targeting expressive phonology for children with developmental speech and language disorder: A systematic review and meta-analysis. *International Journal of Language & Communication Disorders*, *59*(6), 2686–2705. https://doi.org/10.1111/1460-6984.13110

Lin, Y.-S., Tseng, S.-C., & Jang, J.-S. R. (2024). Leveraging phonemic transcription and Whisper toward clinically significant indices for automatic child speech assessment. In *Proceedings of Interspeech 2024*. https://doi.org/10.21437/Interspeech.2024-1887

Maas, E., Robin, D. A., Austermann Hula, S. N., Freedman, S. E., Wulf, G., Ballard, K. J., & Schmidt, R. A. (2008). Principles of motor learning in treatment of motor speech disorders. *American Journal of Speech-Language Pathology, 17*(3), 277–298. https://doi.org/10.1044/1058-0360(2008/025)

Maggu, A. R., Kager, R., To, C. K. S., Kwan, J. S. K., & Wong, P. C. M. (2021). Effect of complexity on speech sound development: Evidence from meta-analysis review of treatment-based studies. *Frontiers in Psychology, 12*, 651900. https://doi.org/10.3389/fpsyg.2021.651900

McLeod, S., Harrison, L. J., & McCormack, J. (2012). The Intelligibility in Context Scale: Validity and reliability of a subjective rating measure. *Journal of Speech, Language, and Hearing Research, 55*(2), 648–656. https://doi.org/10.1044/1092-4388(2011/10-0130)

Meyer, M. K., & Munson, B. (2021). Clinical experience and categorical perception of children's speech. *International Journal of Language & Communication Disorders, 56*(2), 374–388. https://doi.org/10.1111/1460-6984.12610

Namasivayam, A. K., Kent, R., Preston, J. L., Maassen, B. A. M., Hagedorn, C., Nip, I. S. B., McAllister, A., Wang, J., Hustad, K. C., Ménard, L., Bahar, N., Moore, J. G., Petrosov, J., & van Lieshout, P. (2025). Reevaluating the classification of pediatric speech sound disorders: A ground truthing perspective. *Frontiers in Human Neuroscience, 19*, 1700505. https://doi.org/10.3389/fnhum.2025.1700505

Namasivayam, A. K., Pukonen, M., Goshulak, D., Hard, J., Rudzicz, F., Rietveld, T., Maassen, B., Kroll, R., & van Lieshout, P. (2015). Treatment intensity and childhood apraxia of speech. *International Journal of Language & Communication Disorders, 50*(4), 529–546. https://doi.org/10.1111/1460-6984.12154

Peterson, R. L., Pennington, B. F., Shriberg, L. D., & Boada, R. (2009). What influences literacy outcome in children with speech sound disorder? *Journal of Speech, Language, and Hearing Research, 52*(5), 1175–1188. https://doi.org/10.1044/1092-4388(2009/08-0024)

Raitano, N. A., Pennington, B. F., Tunick, R. A., Boada, R., & Shriberg, L. D. (2004). Pre-literacy skills of subgroups of children with speech sound disorders. *Journal of Child Psychology and Psychiatry, 45*(4), 821–835.

Shriberg, L. D., Austin, D., Lewis, B. A., McSweeny, J. L., & Wilson, D. L. (1997). The percentage of consonants correct (PCC) metric: Extensions and reliability data. *Journal of Speech, Language, and Hearing Research, 40*(4), 708–722. https://doi.org/10.1044/jslhr.4004.708

Shriberg, L. D., & Kwiatkowski, J. (1990). Self-monitoring and generalization in preschool speech-delayed children. *Language, Speech, and Hearing Services in Schools, 21*(3), 157–170. https://doi.org/10.1044/0161-1461.2103.157

Silver, S., Fox, A., Gottlieb, M., Siu, M., & Ndungu, J. (2026). *Children's articulation practice in SpeechLP across two months* [Preprint; retrospective observational study]. SpeechLP. https://doi.org/10.5281/zenodo.23932485

Strömbergsson, S., Edlund, E., Pettersson, M., Phan, N., & Kurimo, M. (2026). Self-supervised app-based speech training for children with speech sound disorder: A single-case experimental design study. *International Journal of Language & Communication Disorders, 61*(1), e70163. https://doi.org/10.1111/1460-6984.70163

Sugden, E., Baker, E., Munro, N., & Williams, A. L. (2016). Involvement of parents in intervention for childhood speech sound disorders: A review of the evidence. *International Journal of Language & Communication Disorders, 51*(6), 597–625. https://doi.org/10.1111/1460-6984.12247

Sugden, E., Baker, E., Munro, N., Williams, A. L., & Trivette, C. M. (2018). Service delivery and intervention intensity for phonology-based speech sound disorders. *International Journal of Language & Communication Disorders, 53*(4), 718–734. https://doi.org/10.1111/1460-6984.12399

Tambyraja, S. R. (2020). Facilitating parental involvement in speech therapy for children with speech sound disorders: A survey of speech-language pathologists' practices, perspectives, and strategies. *American Journal of Speech-Language Pathology, 29*(4), 1987–1996. https://doi.org/10.1044/2020_AJSLP-19-00071

Walquist-Sørli, L., Caglar-Ryeng, Ø., Furnes, B., Nergård-Nilssen, T., Donolato, E., & Melby-Lervåg, M. (2025). Are speech sound difficulties risk factors for difficulties in language and reading skills? A systematic review and meta-analysis. *Journal of Speech, Language, and Hearing Research, 68*(1), 164–177. https://doi.org/10.1044/2024_JSLHR-24-00170

This brief was drafted with AI assistance and reviewed by the authors.